Department of Mathematics and Natural Sciences

Physics -112 Lab Assignment: 02

Resources:

Link for online lab:

https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_all.html

List and link for the graph-plotting softwares:

- Desmos (Online): Desmos | Let's learn together.
- Graph (Offline): Download | Graph (padowan.dk)

Tutorials:

- Tutorial link for plotting in Desmos :

https://www.youtube.com/watch?v=-lIUNWVKnUY

- Tutorials link for the Graph :

How to install graph software:

https://youtu.be/e19JqLJMx3A

How to draw a curve using graph software:

https://youtu.be/QBkdzU_8vVo

How to calculate the slope of a line using graph software:

https://youtu.be/z4cMiUFu5j8

Objective:

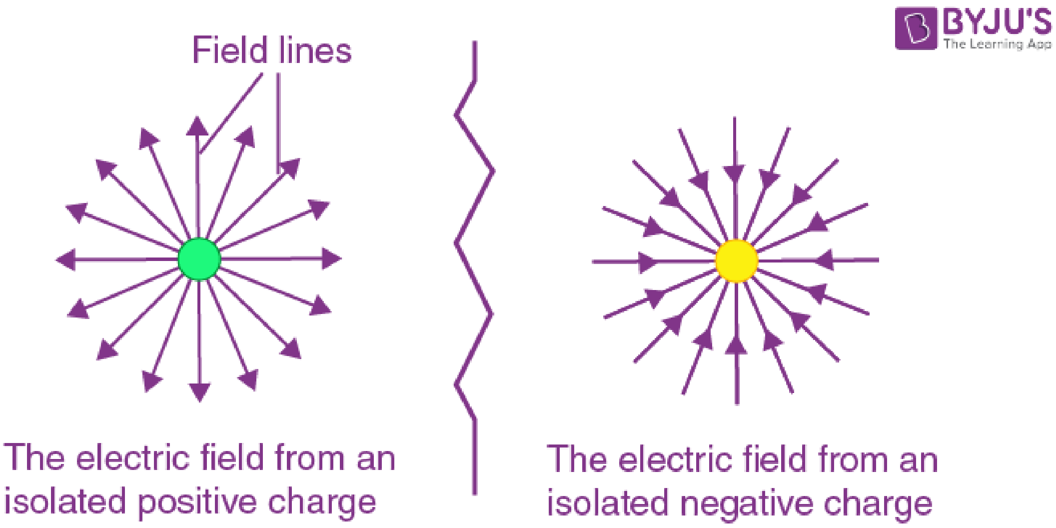
In these activities you will use the Simulation: *Charges and Fields* to develop your understanding of the relationship between electric fields and electric potential.

Theoretical Background:

The Electric field is a vector quantity that exists at every point in space. The electric field at a location indicates the force that *would* act on a unit positive test charge *if* placed at that location.
The electric field is related to the electric force that acts on an arbitrary charge by,

$$\boldsymbol{E = F/q}$$

The dimension of the electric field is newtons/coulomb.



Electric potential energy is the energy that is needed to move a charge against an electric field. You need more energy to move a charge further in the electric field, but also more energy to move it through a stronger electric field. Mathematically we can say that,

$$\mathbf{E = W/Q}$$

The dimension of the electric potential is joules/coulomb.

Activity 1:

Go to the following link : https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_all.html

1. From the box at the bottom of the screen, drag a red **+1 nC** charge into the middle of the screen.

2. If not already selected: Select ‘Electric Field’. How does the brightness of the arrow relate to the strength of the field? What happens when you check/uncheck ‘Direction only’? Which way do the arrows point for a positive charge?

   - The brightness of the arrows likely relates to the strength of the field - brighter arrows indicate a stronger field.
   - When "Direction only" is checked, it shows only the direction without indicating strength.
   - For a positive charge, the arrows should point outward from the charge.

3. Drag the red **+1 nC** charge back into the box at the bottom, and then drag a blue **–1 nC** charge onto the screen. Which way do the electric field arrows point for a negative charge?

   - For a negative charge, the electric field arrows should point inward towards the charge.

4. Click on the yellow Sensor at the bottom and drag it across the electric field. What information do the Sensors show?

   - The Sensor shows information about the electric field strength and direction at different points.

5. What happens to the electric field as you move further from the charges?

    - The electric field strength typically decreases.

6. Take the Voltage meter (labeled '0.0 V'). What information does the voltmeter give? What information is given when you click on the pencil (you should have a green circle)? What does the green circle represent? (If you're not sure, move on and come back to this later.)

    - The Voltage meter (labeled '0.0 V') gives the electric potential at a point.
    - Clicking on the pencil allows to set a reference point (green circle) for voltage measurement.
    - The green circle represents the point where the voltage is being measured in the simulation.

Activity 2:

1. Stack six **+1 nC** charges on top of each other somewhere on the left side of the screen. (It can go anywhere, but there needs to be enough space to measure 8 m away.)

2. From the box at the bottom, drag a Sensor and place it **1 m** to the right of your charge. This sensor measures the E field at the location of its placing. In the table, record the **E** field magnitude at a distance **r** of **1 m**. Ignore the degrees. (*Check / Uncheck Values* as need be)

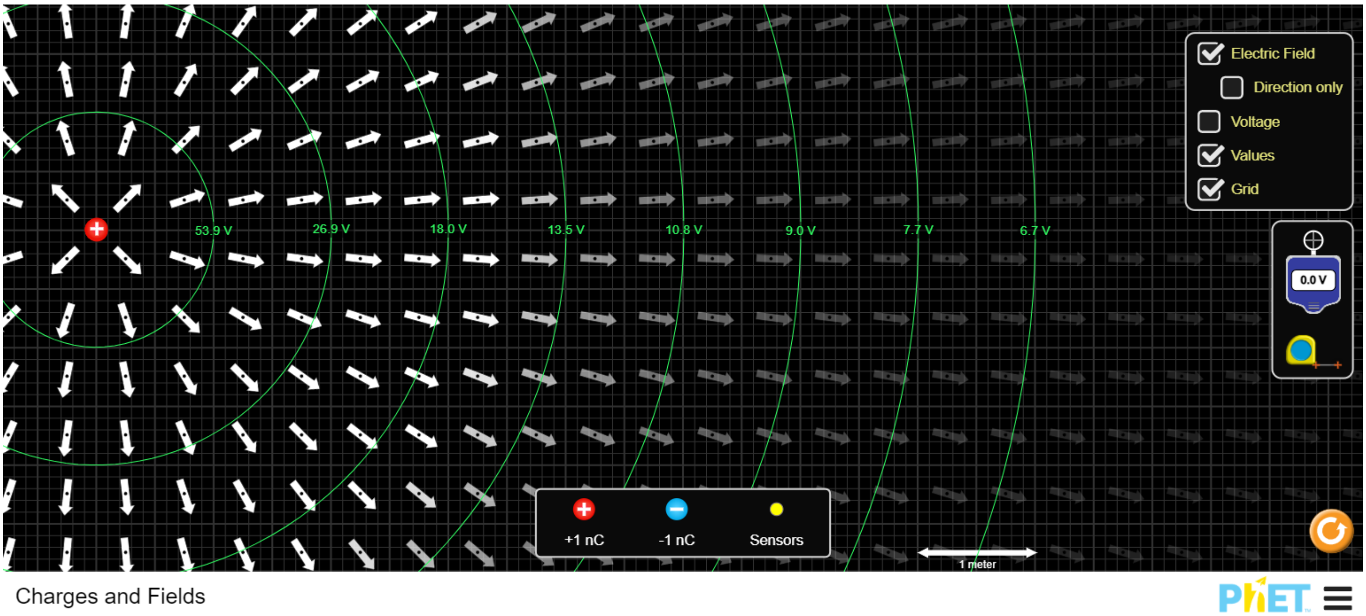
3. Drag the Sensor to the other distances shown in the table, then record the E field measurements.

Distance, r (m)	Electric Field, E (V/m)
1	53.1
2	13.3
3	6

4	3.37
5	2.15
6	1.50
7	1.10
8	0.84

4. Drag your Sensor back and replace it in the box at the bottom of the screen.
5. Using the voltmeter, record the potential *V* by drawing a green circle on the screen at each distance. Fill in the table on the next page. Include a screenshot with all of the green circles at the end of this document.

Distance, r (m)	Voltage, V (V)
1	53.9
2	26.9
3	18
4	13.5
5	10.8
6	9
7	7.7
8	6.7



6. Write the equation for the electric field at any distance *r* from a point charge *q*:

$$E = \frac{kq}{r^2}$$

7. Write the equation for the potential at any distance *r* from a point charge *q*:

$$V = \frac{kq}{r}$$

8. Using the previous table(s), make a graph of electric field *E* and distance *r* to determine Coulomb's constant *k* using the appropriate trendline.

   (Hint: there are 2 ways to do this. Either make a graph and then create the appropriate trendline, or figure out how to make the graph into a straight line and then use a linear trendline. Once you have a trendline, compare the equation written above to the equation of the trendline to find *k*)

➔ The electric field E due to a point charge is given by Coulomb's law:

$$E = kQ\left(\frac{1}{r^2}\right)$$

To make the graph linear, we can plot ($E\ vs\left(\frac{1}{r^2}\right)$) because:

$$E\ \alpha\ \left({}^{1}/{}_{r^2}\right)$$

So, if we plot E on the y-axis and $\left({}^{1}/{}_{r^2}\right)$ on the x-axis, we should get a straight line with a slope equal to kQ.

Distance, r (m)	$^{1}/_{r^2}\ (m^{-2})$	**Electric Field, E (V/m)**
1	1	53.1
2	1/4	13.3
3	1/9	6
4	1/16	3.37
5	1/25	2.15
6	1/36	1.50
7	1/49	1.10
8	1/64	0.84

From the graph we can find 2 points,

$$x_1 = 0.31,\ y_1 = 16.84,$$

$$x_2 = 0.46,\ y_2 = 25.34$$

Slope, m = $\frac{y_2 - y_1}{x_2 - x_1} = \frac{25.34 - 16.84}{0.46 - 0.31}$ = 54.13

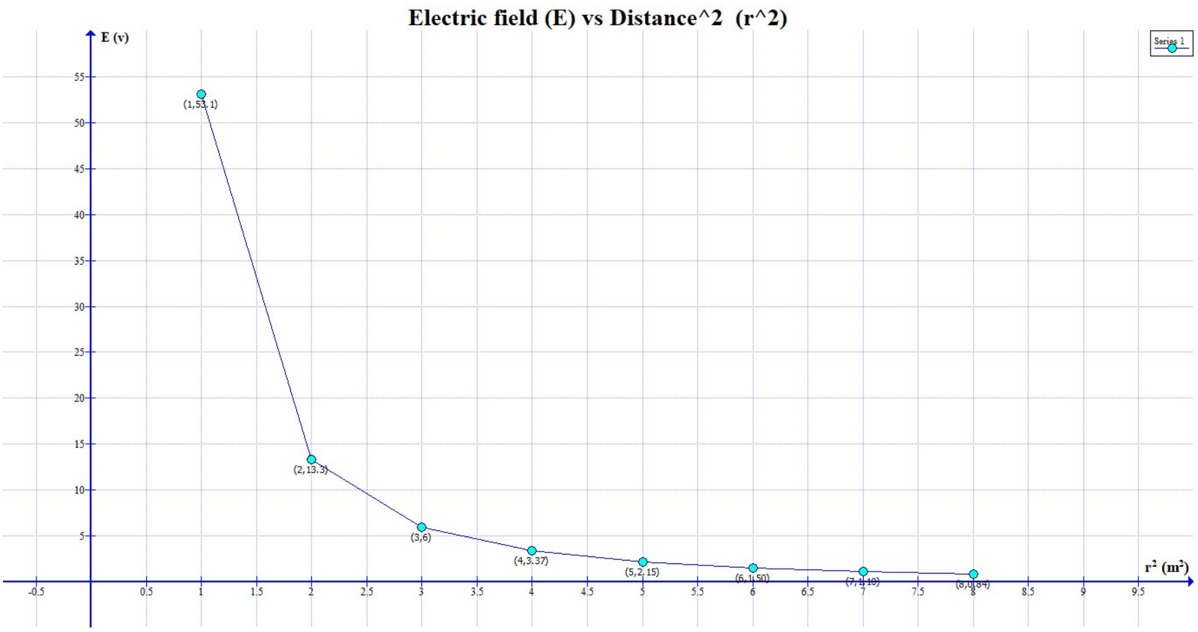
From the equation, we can identify the slope kQ as approximately 54.13

$$kQ = m$$

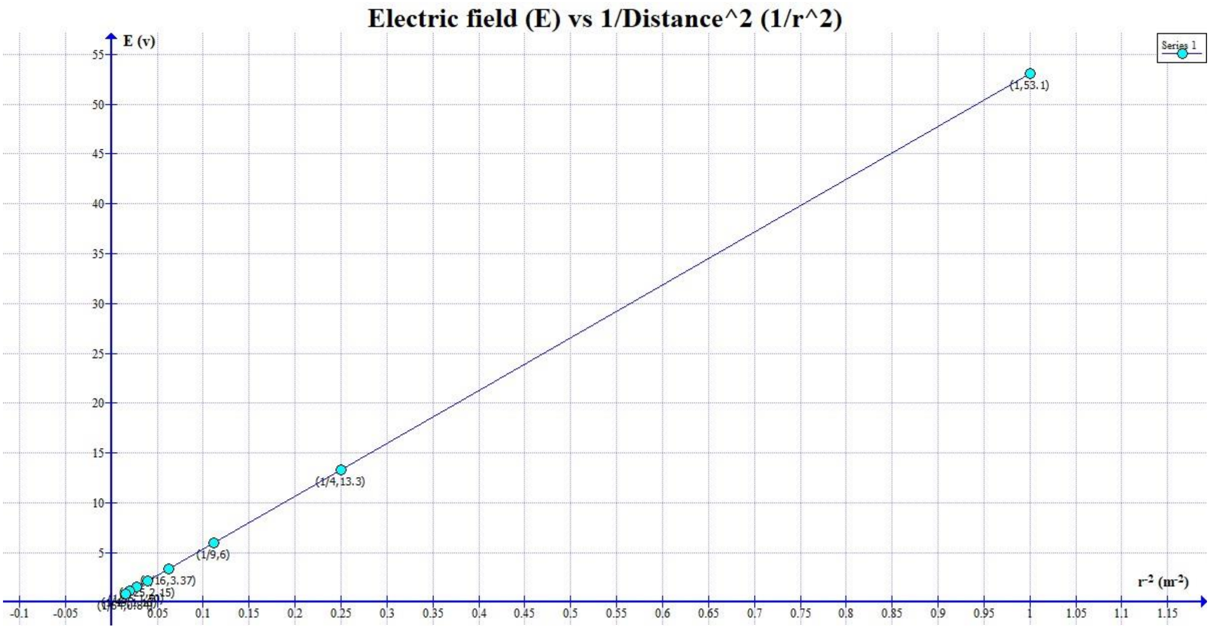
$$=> k = \frac{m}{Q}$$

$$=> k = \frac{54.134}{6 \times 10^9}$$

$$=> k = 9.02 \times 10^9 Nm^2/C^2$$



9. Insert the graph below and write down the *k* value that you found. Compare this value to the known value found on the equation sheet or in class slides using percent error or percent difference (whichever is most appropriate)?



$$Percent\ Error = \frac{|Experimental\ Value - Theoretical\ Value|}{Theoretical\ Value} \times 100\%$$

$$= \frac{|9.02 \times 10^9 - 9 \times 10^9|}{9 \times 10^9} \times 100\%$$

=0.25%

10. Using the previous table(s), make a graph of voltage V and distance r to determine the constant k again using the appropriate trendline.

    (The same hint as above applies, but the work will be slightly different because the equation is different.)

⇨

Distance, r (m)	$^{1}/_{r}\ (m)$	**Voltage, V (V)**
1	1	53.9
2	1/2	26.9
3	1/3	18

4	1/4	13.5
5	1/5	10.8
6	1/6	9
7	1/7	7.7
8	1/8	6.7

We know that,

$$V = \frac{kQ}{r}$$

Here, to make the graph linear, we can plot ($V\ vs \left(\frac{1}{r}\right)$) because:

$$V\ \alpha\ \left({}^{1}/_{r}\right)$$

From the graph we can find 2 points,

$$x_1 = 0.39,\ y_1 = 21.27,$$

$$x_2 = 0.48,\ y_2 = 26.04$$

Slope, m = $\frac{y_2 - y_1}{x_2 - x_1} = \frac{26.04 - 21.27}{0.48 - 0.39}$ = 53.9

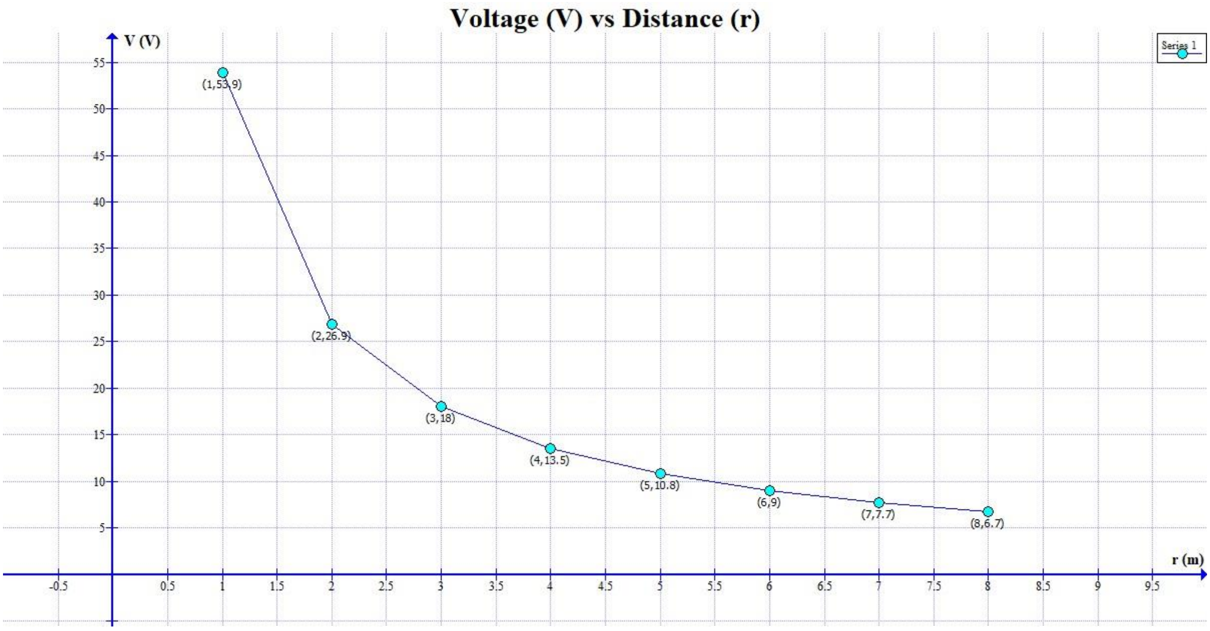
From the linear regression, we found the slope of the line to be approximately 53.9. This slope corresponds to kQ in the equation, $V = \frac{kQ}{r}$

$$kQ = m$$

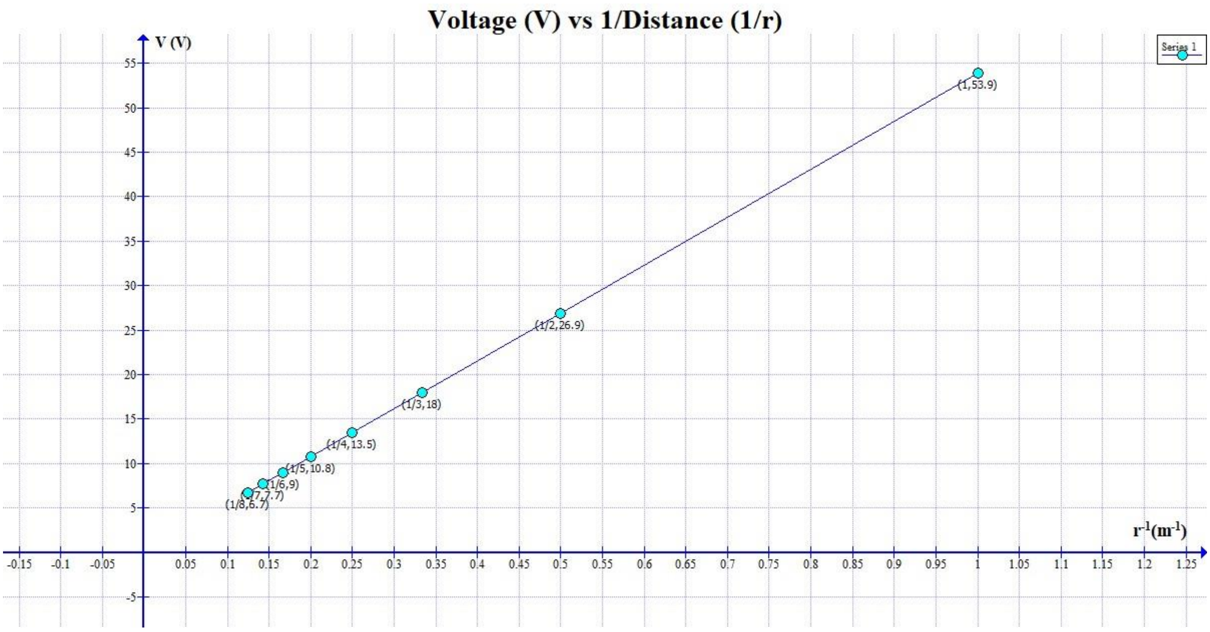
$$=> k = \frac{m}{Q}$$

$$=> k = \frac{53.9}{6 \times 10^{-9}}$$

$$=> k = 8.98 \times 10^9 Nm^2/C^2$$



11. Insert the graph below and write down the k value that you found. Compare this value to the known value using percent error/difference?



$$Percent\ Error = \frac{|Experimental\ Value - Theoretical\ Value|}{Theoretical\ Value} \times 100\%$$

$$= \frac{|9 \times 10^9 - 8.98 \times 10^9|}{9 \times 10^9} \times 100\%$$

$$= 0.19\%$$